The Triaxial Polymer Ejection System

Polymer:

4g Silk Fibroin

1g Borate buffer

3g Tannic Acid

3g Chitosan

Note:

These ingredients are mixed in 89mL of water to create an **11%(w/w) concentrated solution.**
Disclaimer: The amounts of these chemicals are speculative and are not yet proved.

Procedure to obtain Silk Fibroin:

Silk fibroin is manually derived from **silk cocoons**.
The **silk cocoons** are boiled in a solution of **baking soda** and **water**.
The obtained **silk strands** are converted to a liquid using **Ajisawa's reagent**.

The resulting compounds will be **pure lab grade silk fibroin**.

Pressure chamber:

The liquid polymer is routed through a pressure chamber made of **carbon fiber** and reinforced with an **outer titanium shell**.
When the polymer enters the pressure chamber it is **sealed automatically on both exits**.

CO2 is released into the chamber **increasing the pressure** inside relative to the outside.

When the seal at the exit is **released**, the fluid will rush out as pressure becomes **equal**. The **maximum pressure** at its peak will be **~100 PSI.**

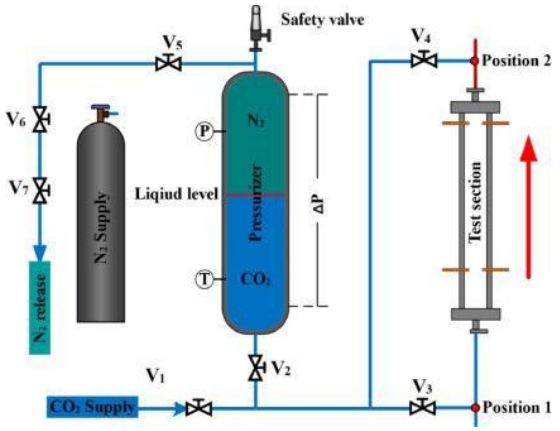


Fig 1.1 –CO2 based Pressurizer

Nozzle:

The mechanism utilizes a **triaxial nozzle** where the inner channel is **20-gauge**, and the outer channel is **12-gauge**. The middle channel will be **16-gauge** for optimal release velocities.

Inner channel: polymer fluid

Middle channel: 100% pure acetone

Outer channel: CO2 (compressed)

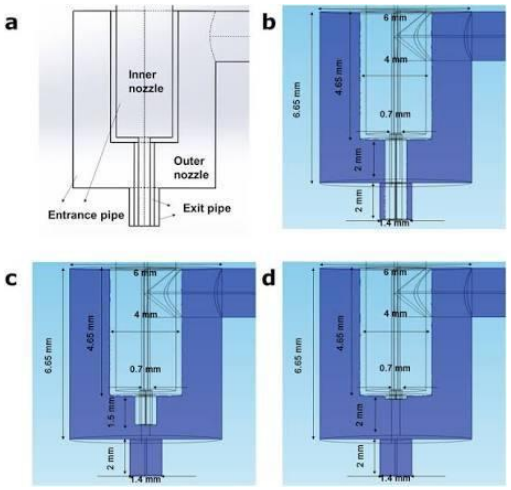


Fig 1.2 – Triaxial Nozzle

Reasoning for the CO2:

The compressed CO2 will help in **strand alignment**; the CO2 also exploits the **venturi effect** which improves the range of the strand.

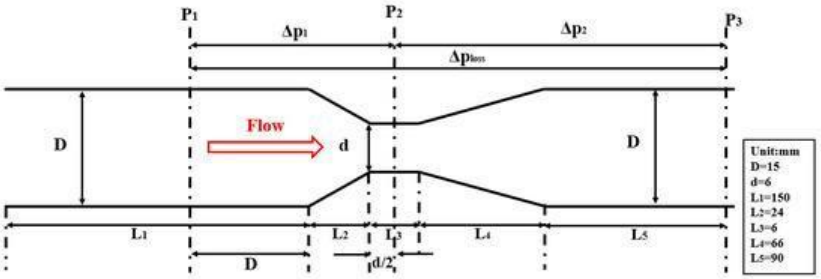


Fig 1.3 – Venturi Effect

Chemistry behind the polymer:

The polymer uses **chitosan** in the mixture to increase the **load bearing capacity** up to **800x its weight**.

The polymer also utilizes **tannic acid/pyrocatechol** as its **main adhesive**. Derived from **mussel biochemistry**.

The **borate buffer** compliments the adhesive and increases its ability to stick by **18-fold**.

Tubing:

The system utilizes **Teflon tubing** to **transport substances** to the **chambers safely**.

Storage:

The substances are stored inside **HDPE tanks** that are to be **worn on the back**.

Transportation:

The system utilizes **peristaltic pumps** at **specific junctions** along the **Teflon pipes** to **transport liquids** across the tubing and into **the compressor**.

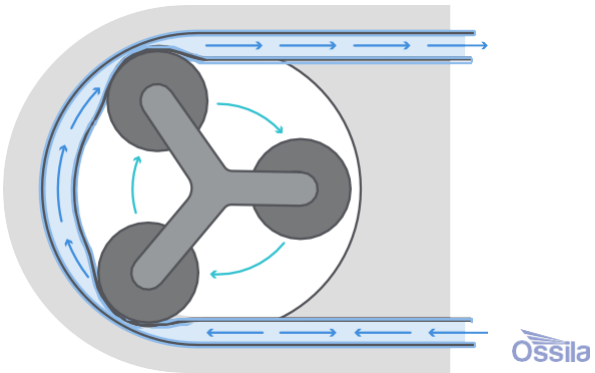


Fig 1.5 – Peristaltic Pump

Release mechanism:

The system **detects for lever presses constantly**. If the user were to **pull the lever up**, the **substances are deployed** from their **respective containers** to their **respective pressure chambers**, CO2 is released and the system is on standby. If it detects a **lever press**, the **seal is released** and the **fluid is ejected** until the **lever is released**. When a **second press** is **detected**, the **strand** is **cut**. The **subsequent press** will **start a new strand**. If the **compressor is empty**, an onboard **speaker will play an audio message** stating that the **system must be reloaded**.

Note 1:

The formula used for this mechanism is inspired by and derived from the **Tuft's university Boston's silk lab experiment in 2024**.

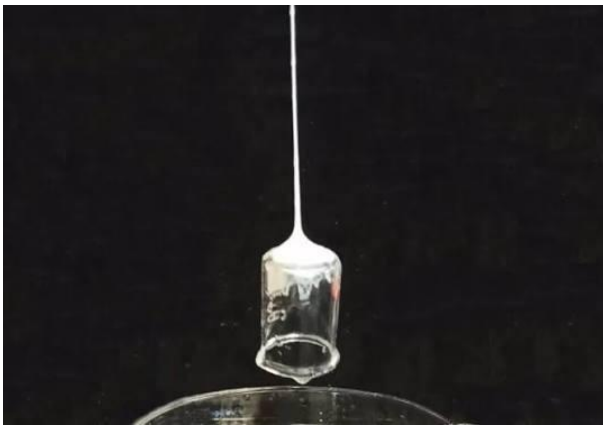


Fig 1.4 – Tufts University SilkLab Experiment

Note 2:

To prevent **catastrophic backflow**, when the user **releases** the **lever**, the **inner channel** shuts down **first**, the **middle channel** shuts down **1ms later** and the **CO2 shuts down 200ms later**.

Note 3:

The original chemical 'recipe' calls for Dopamine as it belongs to the catechol group. Though Dopamine isn't freely available to the general public, therefore it has been swapped with Tannic acid. Pyrocatechol was taken into consideration but was later deemed unfit for poor reactive capabilities and difficulty to maintain pH balance suitable for borate buffer to stay dissolved in water.

Images from:

- MDPI
- ScienceDirect
- ResearchGate
- Tufts Now
- Ossila